

Dynamic Geometric Quantiles with DGQ

1 What DGQ computes

Given a collection of N multivariate time series, each a trajectory $X_i = (X_i(1), \dots, X_i(T))$ in $\mathbb{R}^d$, the Dynamic Geometric Quantile picks out a quantile-like trajectory for a chosen tilt direction u in the open unit ball. The single function `DGQ()` returns two related objects:

- the **empirical DGQ** (*constrained*): a real observed series, the one minimizing $C(j) + N\langle u, V(j) \rangle$; and
- the **time-wise geometric quantile** (*unconstrained*): a free trajectory whose value at each time is Chaudhuri's geometric u -quantile of that time's cross-section.

The centre $u = 0$ gives the dynamic median (the ρ -medoid). Tilting u moves the represented quantile in a chosen direction while the constrained version stays a genuine, coherent observed path.

```
library(DGQ)
```

2 A small collection

We build $N = 200$ two-dimensional series, each a smooth oscillation around a cross-sectional offset, so the trajectories cross and the time-wise quantile jumps between curves while a single observed series stays coherent.

```
set.seed(1)
N <- 200; T <- 60; d <- 2
tt <- seq_len(T) / T
offsets <- cbind(rnorm(N, sd = 2), rnorm(N, sd = 1.4))
X <- array(0, dim = c(N, T, d))
for (i in seq_len(N)) {
  f1 <- runif(1, 1, 3); f2 <- runif(1, 1, 3)
  a <- runif(1, 0.8, 1.6)
  osc <- cbind(a * sin(2 * pi * f1 * tt), a * cos(2 * pi * f2 * tt))
  X[i, , ] <- sweep(osc + matrix(rnorm(T * d, sd = 0.15), T, d),
                        2, offsets[i, ], "+")
}
```

3 The dynamic median

```
fit0 <- DGQ(X, u = c(1, 0)) # u = c(1, 0), pooled-whitening metric (the default)
fit0
```

```
#> Dynamic Geometric Quantile (DGQ)
#> collection : N = 200 series, T = 60 times, d = 2 variables
#> metric      : pooled      depth: exact      cumulative: no
#> medoid      : series #128 (dynamic median, u = 0)
#> empirical DGQ (constrained, observed series) by direction:
#> tau=0.500    -> series #128
#> time-wise geometric quantile (unconstrained): 1 direction(s)
```

4 Tilted directional DGQs

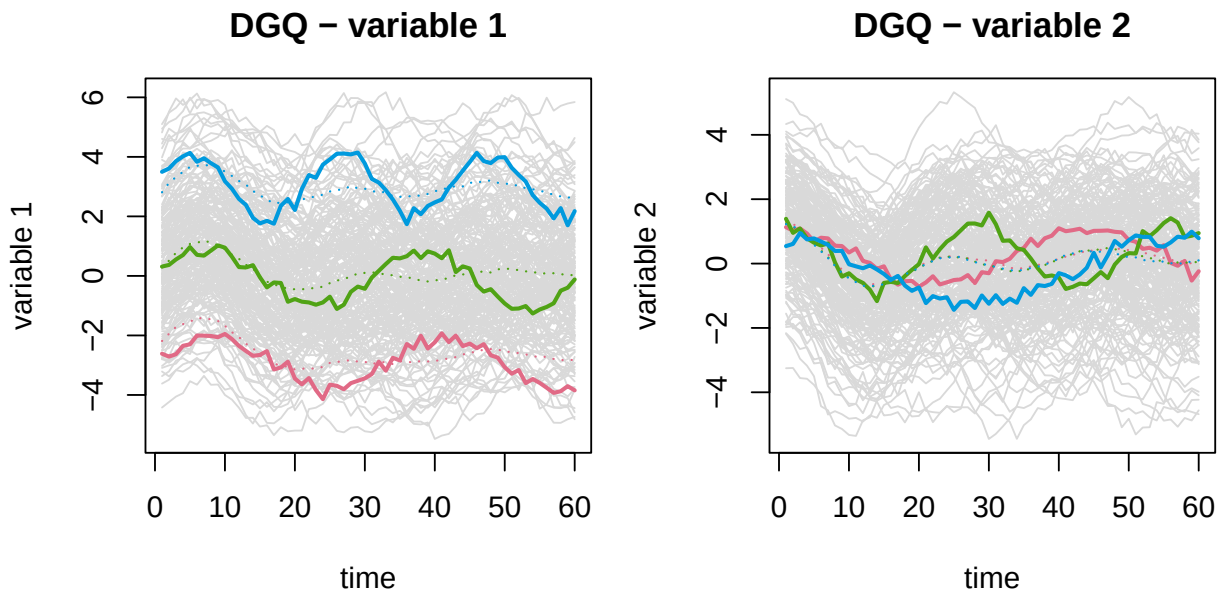
Supply a direction u and a vector of quantile levels τ (default 0.5, representing the median). The tilt directions are computed as $(2 * \tau - 1) * u_norm$.

```
fit <- DGQ(X, u = c(1, 0), tau = c(0.15, 0.5, 0.85))
fit$empirical$index           # observed series representing each direction

#> tau=0.150 tau=0.500 tau=0.850
#>      184      128      11
```

The empirical trajectories (solid) are real observed paths; the time-wise geometric quantiles (dashed) are the unconstrained targets, visibly damped because they average across crossing curves.

```
plot(fit)
```

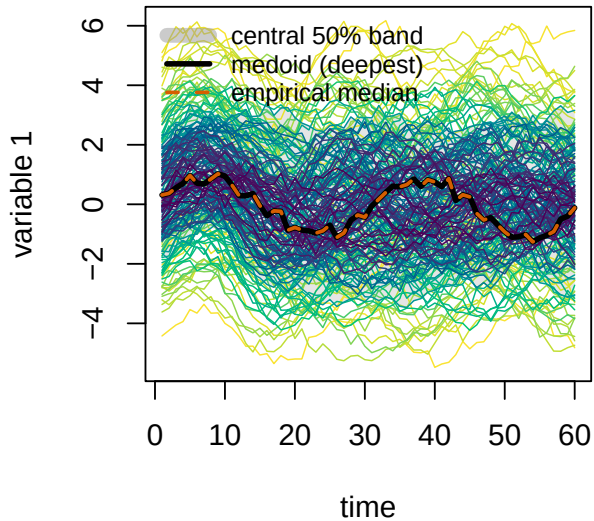


Beyond the default trajectory overlay (`type = "trajectory"`), `plot()` offers two further views of the selected series through the `type` argument.

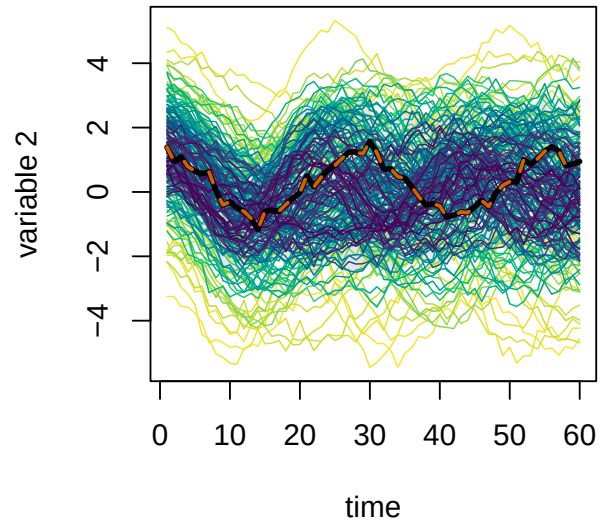
The "rainbow" view colours every series by its trajectory depth $C(j)$ —deep, central series dark and shallow, outlying series bright—shades the central-50% band, and overlays the dynamic medoid together with the empirical median (drawn when a $\tau = 0.5$ direction is present).

```
plot(fit, type = "rainbow")
```

DGQ rainbow – variable 1



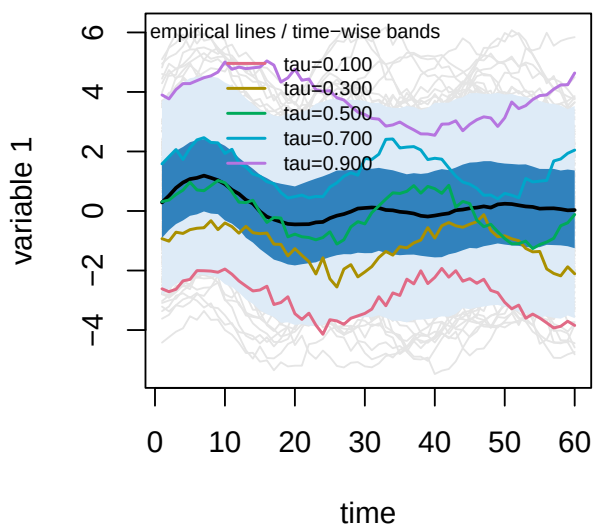
DGQ rainbow – variable 2



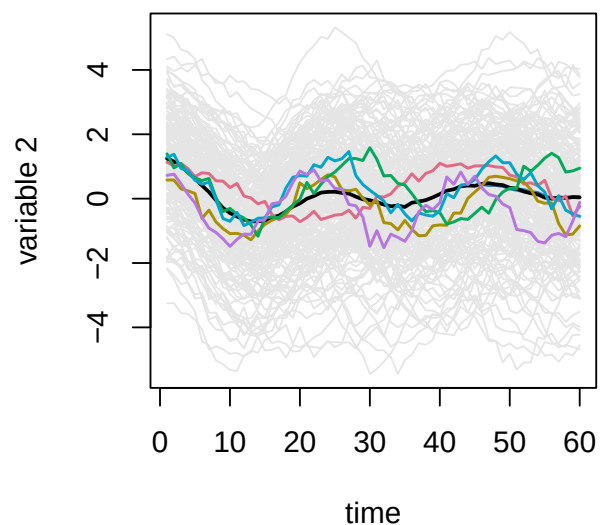
The "fan" view draws the time-wise quantile fan—nested bands between symmetric τ pairs plus the median—and overlays the empirical per- τ selections as lines. Because the empirical DGQs are real observed series, their lines may cross or coincide, whereas the time-wise bands are smooth and nested. This view needs a fit with at least two quantile levels.

```
fit_fan <- DGQ(X, u = c(1, 0), tau = c(0.1, 0.3, 0.5, 0.7, 0.9))
plot(fit_fan, type = "fan")
```

DGQ fan – variable 1



DGQ fan – variable 2



5 Metrics, depth, and input formats

- `metric = "pooled"` (default) whitens once for exact affine invariance; `metric = "timewise"` whitens with the per-time covariance; `metric = "none"` uses raw Euclidean geometry.
- `method = "clara"` replaces the exact $O(N^2Td)$ depth with a CLARA subsample estimate, which is cheaper for large N :
- `n.cores` requests OpenMP cores for empirical exact or CLARA candidate evaluation. The default `n.cores = 1L` is serial; builds without OpenMP support also run serially.

```
fit_clara <- DGQ(X, u = c(1, 0), tau = c(0.15, 0.5, 0.85), method = "clara", sample.size = 80)
fit_clara$empirical$index

#> tau=0.150 tau=0.500 tau=0.850
#>          35          148          179
```

Input may be a three-dimensional array $[N, T, d]$ or a list of $T \times d$ matrices:

```
Xl <- lapply(seq_len(N), function(i) X[i, , ])
identical(DGQ(Xl, u = c(1, 0))$medoid, DGQ(X, u = c(1, 0))$medoid)

#> [1] TRUE
```

6 Cumulative series

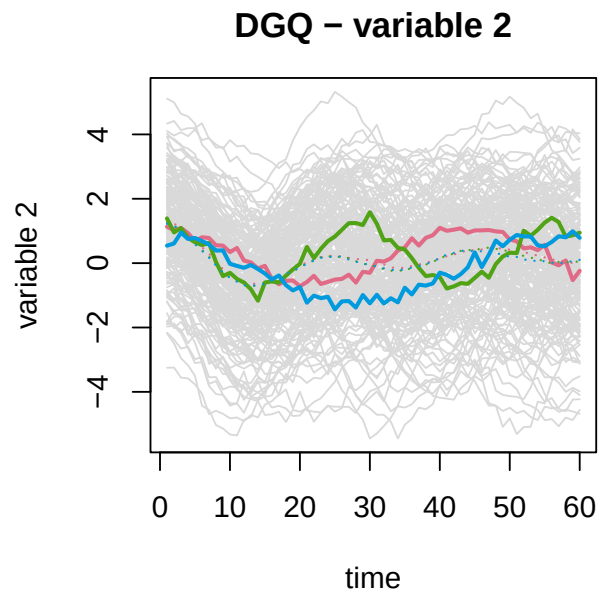
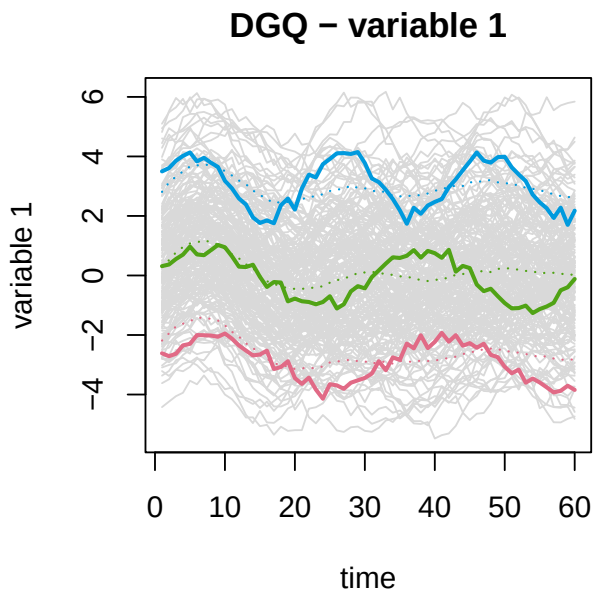
For increment or flow data, set `cumulative = TRUE` to analyze cumulative levels. Each series-variable path is cumulatively summed over time before whitening and DGQ computation, so the geometry, the depth $C(j)$, and both DGQ branches operate on the integrated levels; the returned trajectories and every `plot()` view are then in cumulative units. This matters when the raw data are increments (returns, flows, rainfall) but the quantity of interest is the accumulated total: the dynamic median then represents the most central *accumulation path* rather than the most central oscillation.

```
fit_cumulative <- DGQ(X, u = c(1, 0), tau = c(0.15, 0.5, 0.85), cumulative = TRUE)
fit_cumulative$cumulative

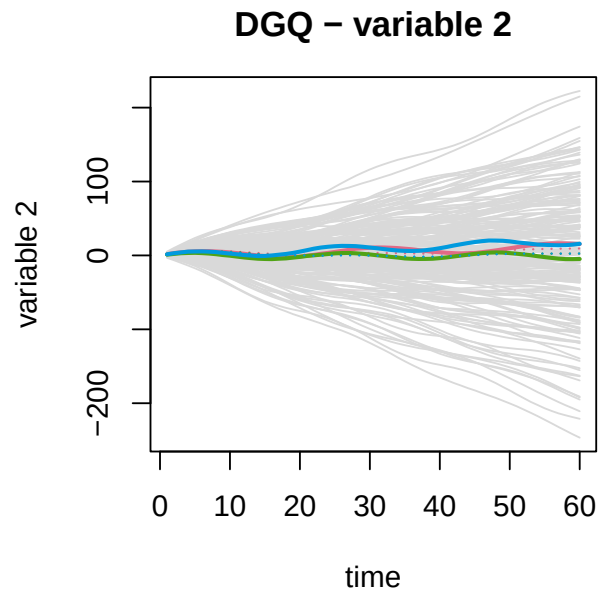
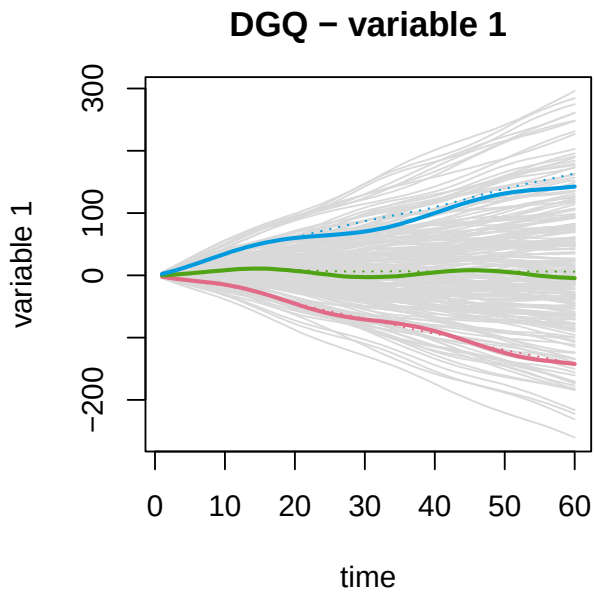
#> [1] TRUE
```

The contrast is clearest side by side. On the increments the trajectories oscillate around their offsets and the time-wise quantile is heavily damped; on the cumulative levels the paths spread into trending fans and the empirical DGQ tracks a coherent accumulation path.

```
plot(fit) # increments (Section 4 fit)
```



```
plot(fit_cumulative) # cumulative levels
```



The depth-ranked view makes the same point: under accumulation, central and outlying paths separate more strongly because early differences persist.

```
plot(fit_cumulative, type = "rainbow")
```

